Structure and Data: QCA as a Framework for Separating Emergent Structure and Contingent Parameters in Particle Physics

Sergey A. Inozemtsev

Version 14.4 (Final) · November 11, 2026

Abstract

Central thesis: Particle physics splits into two distinct classes — *structure*, which can be derived from the local unitary rules of a quantum cellular automaton (QCA), and *data* (parameters), which must be assigned externally. We test this thesis numerically across 19 Jupyter notebooks. The *structure* — encompassing the Dirac equation, gauge symmetries, the Higgs mechanism, and three fermion generations — is reproduced from QCA rules with an accuracy of 10^{-15}–10^{-17} (8 independent derivations, Class A). Consistency checks with known quantum mechanical theorems (CPT, $c = 1$, entanglement, Pauli exclusion; Class B) and negative controls (square lattice, fundamental constants, CP conservation in 1D and 2 generations; Class C) define the boundary of this classification. Conversely, *parameters* (masses, Yukawa couplings, fundamental constants, CP phase) cannot be derived and constitute input data (4 data classifications). Additionally, we investigate the hypothesis of a base-15 structure for lepton masses: using pairwise logarithmic ratios m_j/m_i (which are scale-invariant by construction), base 15 yields a deviation of 0.028148 for leptons. A permutation test over 84 partitions of the 9 masses gives $p_{abs} = 10/84 = 0.1190$; the status is **not established** (deficit to the suggestive threshold $5/84 = 0.0595$). The look-elsewhere effect is prominent: the triplet $\{\mu, b, t\}$ at base 40 yields a deviation of 0.005912, which is 4.76 times better than the lepton signal. For quarks, no base yields a deviation below 5% (MC mean at base 40 = 0.178335; deterministic $\text{dev}_{\text{best39}} = 0.174249$). No new predictions are made; the measured value $\Sigma m_\nu = 0.0608$ eV (nb13) is consistent with the DESI DR2 upper bound — this is a consistency check, not a prediction. The framework does not depend on the base-15 hypothesis.

1 Introduction

1.1 The Epistemological Paradox

The project began with a question about artificial intelligence. There exists a two-way epistemological wall between us and AI. The inversion of this observation is: *what if we ourselves are in the position of an AI relative to some substrate?* We see the "output" (observable physics), but we do not see the "hardware" (the substrate).

1.2 Analogy: "Arithmetic and Ink"

Imagine a sheet of paper with numbers written on it. You are a mathematician who knows only arithmetic and geometry. You can add numbers and measure the distances between them. But you **cannot** deduce from arithmetic why the digit "7" is drawn with straight lines while "2" is curved, or why "5" is written in red and "7" in blue. These properties are **real**, but they are

not part of the mathematical rules. They are **characteristics of the medium** (the ink, the handwriting).

In this project, we show that physics is structured analogously: **rules** (symmetries, interactions, the speed of light, CPT, entanglement, three generations) are derivable, whereas **parameters** (masses, coupling constants, the CP phase) are the "ink" that cannot be derived from the rules.

1.3 What This Approach Does *Not* Claim

We emphasize three points to prevent misunderstanding.

First, the approach does **not** claim that the Universe is a *simulation*. There is no "simulator outside" executing our world as a program. QCA is an **ontology**, not the simulation hypothesis: the claim is that physics *is* a local unitary rule, not that someone *computes* it. There is no "before the rule" and no "outside the rule" — there is only the rule. This distinction aligns with the information-theoretic view of the universe [9], but it strictly avoids the implication of an external simulator.

Second, the approach does **not** claim to be a "theory of everything". It classifies what is derivable from local unitarity and what must be given as input; it does not derive the Standard Model, gravity, or cosmology. Eight sectors of Class A, four checks of Class B, five negative controls of Class C, and four data classifications — this is the **full scope** of the work, nothing more.

Third, the approach does **not** claim that the substrate is directly knowable. We see the "output" (physics) and classify it into structure and data. We do not see the "hardware". Whether the hardware exists, what it is, and how our output relates to it — the approach **leaves in brackets** and does not resolve. The structure/data classification works *inside* the output, regardless of what lies behind it.

2 Methodology

Quantum Cellular Automaton (QCA) is a minimal model of local unitary computation: a discrete lattice, discrete time steps, and a local rule $\mathcal{U}$. It represents the smallest set of assumptions required to ask the question: "What follows from locality and unitarity?" QCA does not claim ontological fundamentality; it is a minimal testbed. All results are numerical and are verified in `float64` precision.

3 Structure Sectors (Class A)

There are **8 independent derivations** of structure from local QCA rules:

- **nb01:** 1D Dirac equation (10^{-15});
- **nb02:** U(1) gauge field / photon (10^{-17});
- **nb03:** SU(2) / W, Z bosons (10^{-16});
- **nb04:** Independence of gauge invariance and unitarity from mass ($m \in [0.01, 1.5]$; scatter of residuals and variance below the required order);
- **nb05:** Higgs mechanism via a BCS condensate (0.1%);
- **nb06:** 2D Dirac equation on a honeycomb lattice ($\alpha = 1.01$, isotropy 1.0000);
- **nb08:** 3D Dirac on a 4D domain wall ($E_0 = 0$, wall weight 51%);

- **nb18 Parts B–C:** Three generations — N walls $\to$ N modes in 1D SSH (exact correspondence, $|\langle\gamma_5\rangle| = 1.0000$); 2D is realized via overlap fermions (Ginsparg–Wilson residual $\sim 3 \times 10^{-15}$).

3.1 Consistency Checks (Class B)

These are not "confirmation levels" in the same sense — QCA reproduces known quantum mechanical theorems without adding independent content:

- **nb12:** CPT, $m = \bar{m}$;
- **nb15:** $c = 1$ from locality (group velocity $v_F = 1.000000$ in step/tick units);
- **nb16:** CHSH $= 2\sqrt{2}$ is preserved;
- **nb17:** Exchange sign -1 for fermions, $+1$ for bosons.

3.2 Negative Controls (Class C)

- **nb07:** Square 2D lattice yields anisotropy > 1.15 (contrast with the honeycomb lattice);
- **nb11:** Null result for the 4D wall geometry (no hidden 1/32 factors in velocities);
- **nb14:** No hidden grid for fundamental constants (error $\sim 10^{-3}$);
- **nb18 Part A:** Naive 2D lattice — 4 Dirac points (fermion doubling);
- **nb19 T0–T2:** CP is conserved in 1D and 2 generations.

How negative controls were tested. All controls use the same metric dev and the same 5% threshold as the leptons. The 84-triplet permutation test is applied exclusively to the lepton triplet — the controls are evaluated separately using the same metric but are not included in the 84 partitions. Specifically, these controls exclude: (a) **quarks** — demonstrating that the base-15 preference is not a universal property of any mass triplet (0 bases with dev $< 5\%$); (b) **nb14 (constants)** — showing that the preference is not a consequence of metric choice (no hidden grid at 10^{-3}); (c) **nb07 (square lattice)** — proving that the 2D effect is not an artifact of the honeycomb structure (anisotropy > 1.15); (d) **nb18 Part A (naive lattice)** — showing that chiral anomalies do not create an illusion of topology (4 Dirac points explicitly).

3.3 Data Classifications

- **nb09:** SVD of quark masses (rank-1, 99.7%, factor 2 — without a derivable basis);
- **nb10:** Base scan (see Section 7);
- **nb13:** Neutrinos — the measured mass sum $\Sigma m_\nu = 0.0608$ eV is consistent with the DESI DR2 upper bound. This is a **consistency check, not a prediction**: the model does not predict the value in advance and does not fit the parameter. There is no full base-15 grid for neutrinos.
- **nb19 T4–T5:** CP violation — the phase δ cannot be derived from the rules and is an input (see Section 6).

4 Synthesis: 8 + 4 + 5 + 4

The counters do not overlap; negative controls are not counted as confirmation levels. nb18 is split: Part A — Class C, Parts B–C — Class A. nb19 is split: T0–T2 — Class C, T4–T5 — Data.

Class	Count	Notebooks	Role
A — Independent structural derivations	**8**	nb01–06, 08, 18 Parts B–C	Structure is derived
B — Consistency with quantum mechanics	**4**	nb12, 15–17	Consistent with QM theorems
C — Negative controls	**5**	nb07, 11, 14, 18 Part A, 19 T0–T2	Classification is non-trivial
Data — Parameter classifications	**4**	nb09, 10, 13, 19 T4–T5	Remains as input

Table 1: The complete scope of the framework.

5 Three Generations: A Closed Sector (nb18)

5.1 Mechanism and Number

The Jackiw–Rossi mechanism [3] is **exact in 1D**: N SSH domain walls [4] yield exactly N zero modes with $E = 0$ and $|\langle\gamma_5\rangle| = 1.0000$. It is **realizable in 2D** via overlap fermions [2] (Ginsparg–Wilson residual $\sim 3 \times 10^{-15}$). A naive 2D lattice suffers from fermion doubling (nb18 Part A) — this is a lattice artifact, not physics.

The specific number $N = 3$ is an **input fact**: the mechanism allows any N, and the choice $N = 3$ in nature is a parameter.

5.2 Verdict

Three generations: mechanism = structure, number $N = 3$ = input. The mechanism is derived exactly in 1D and is realizable in 2D. This does not depend on the base-15 hypothesis.

6 CP Violation: A Closed Sector (nb19)

T0–T2: CP is conserved in 1D with real and complex masses, as well as in 2 generations (the phase is eliminated by a chiral rotation). T4: 3 generations with a toy CKM matrix — $J(\delta) \neq 0$, maximum at $\delta = \pi/2$, $J = 0$ at $\delta = 0$. T5: Majorana phases yield a non-zero J_M, but they manifest in lepton-number-violating processes ($0\nu\beta\beta$), not in oscillations.

CP violation = data. The framework can accommodate CP violation (via 3 generations or Majorana masses), but the phase δ is not derived from the rules.

7 Base 15: An Exploratory Observation (nb10)

7.1 Statistic

In nb10, we use the statistic of pairwise logarithmic ratios:

$$\mathrm{dev}(b) = \mathrm{mean}_{i<j} \left|\log_b(m_j/m_i) - \mathrm{round}(\log_b(m_j/m_i))\right| . \tag{1}$$

The statistic is scale-invariant by construction. The range of bases is $b \in [2, 40]$.

7.2 Result for Leptons

Base 15 is the best for the lepton triplet: $\mathrm{dev} = 0.028148 \pm 1.61 \times 10^{-5}$ (Monte Carlo over PDG errors [5], seed 42, $N = 500$ — a stability test). It is the only base with a deviation below 5%.

7.3 Permutation Test (Best-of-39)

We consider all 84 partitions of the 9 masses into groups of $3+6$. For each group T, we compute $\mathrm{dev}_{\mathrm{best39}}(T) = \min_{b\in[2,40]} \mathrm{dev}(T,b)$. Then:

$$p_{abs} = \frac{\#\{T : \mathrm{dev}_{\mathrm{best39}}(T) \leq \mathrm{dev}_{\mathrm{best39}}(\mathrm{leptons})\}}{84}. \tag{2}$$

Result: $p_{abs} = 10/84 = 0.1190$. Status: **not established** (deficit to the suggestive threshold $5/84 = 0.0595$).

Note for the reader: The lepton triplet is not extreme in either counter. In the p_{abs} counter (by $\mathrm{dev}_{\mathrm{best39}}$), it ranks 10th out of 84: 9 triplets have a strictly smaller $\mathrm{dev}_{\mathrm{best39}}$, including $\{\mu, b, t\}$ with dev = 0.005912 (§7.4). In the asymmetry counter (by $\mathrm{dev}_6 - \mathrm{dev}_3$), it ranks 16th out of 84.

What was pre-specified. The pre-specified hypothesis was not "base 15 is special", but "some base $b \in [2, 40]$ yields an anomalously small dev". Base 15 turned out to be the minimum in this scan. The metric $p_{abs} = 10/84$ accounts for this search by construction: each triplet is allowed to choose its best base (best-of-39), which is why p_{abs} is 0.1190 rather than $1/84$. Pre-registration is documented in `nb10_pvalue.md` §1–§2 (timestamp 2026-11-11 03:40) before the run.

7.4 Look-Elsewhere Effect

The most "grid-like" triplet under best-of-39 is $\{\mu, b, t\}$ at base 40 with a deviation of 0.005912, which is **4.76 times** better than the lepton signal (0.028148). This is a direct consequence of the look-elsewhere effect: scanning over 39 bases allows random triplets to "try" all bases.

7.5 Quarks

For quarks: MC mean at base 40 = 0.178335 ± 0.0146; deterministic minimum $\mathrm{dev}_{\mathrm{best39}} = 0.174249$. No base yields a deviation below 5%. The best base 40 is unstable (the top-5 lie within ±std of the Monte Carlo over PDG errors: spread 0.0035 < std 0.0146). The negative control is clean.

7.6 Fixed Base 15 (Secondary Line, with Caveat)

If base 15 is fixed without multiplicity correction, $p_{\mathrm{fixed}} = 1/84 = 0.0119$. **Caveat:** base 15 was found by scanning these same data, so p_{fixed} is invalidated by post-selection. It is not used in the abstract or the bottom line.

7.7 Relation to Koide

The Koide relation $K = 2/3$ [1] has been known since 1981 and is not a result of this project. The nearest "nice" fractions (1/32, 1/91) are post-hoc: for any measured value, close fractions can always be found. The real fact of 10^{-5} precision is Koide for measured masses ($K = 0.66666051$, deviation 6.2×10^{-6}).

The 2.8% and Koide are not independent: integer exponents (0, 2, 3) yield $K = 0.65591603$; fractional parts (1/32, 1/91) yield $K = 0.66666378$ — exactly the degree of freedom needed to reach $K = 2/3$. The agreement "2.8% + Koide" is one piece of evidence in two formulations, not two independent ones.

7.8 Monte Carlo over PDG Errors

The Monte Carlo over PDG uncertainties (seed 42, $N = 500$) is a **stability test against measurement errors**, not a test of statistical significance. Result: quarks 0.178335 ± 0.0146, leptons $0.028148 \pm 1.61 \times 10^{-5}$. Significance is determined exclusively by the permutation test.

On the std at bases 16 and 4: the range 10^{-9}–10^{-8} is confirmed by the actual run (5.24×10^{-9} and 1.05×10^{-8}). The reason is structural: at these bases, the contribution of the mass perturbation to the three pairwise terms cancels pairwise to first order, unlike base 15, where they align.

Status of base 15: exploratory, not established. The framework does not depend on this hypothesis. Negative controls are preserved. The primary criterion (best-of-39) is not established ($p_{abs} = 10/84 = 0.1190$); the secondary criterion (fixed base 15, $1/84 = 0.0119$) is only valid with the post-selection caveat.

8 Falsification Criterion

The thesis of the separation of structure and data would be falsified if:

1. A statistically significant base is found for quarks in $b \in [2, 40]$ using the same method as for leptons ($p < 0.01$);
2. Adding local unitary interactions (a) changes the classification of any sector from "structure" to "data", or (b) requires fitting a parameter previously reproduced without fitting;
3. The CP phase is derived from QCA rules without free parameters, which would move it from "data" to "structure".

None of the three has occurred. Base 15 for leptons does **not** pass criterion (i) at $p < 0.01$ — its $p_{abs} = 0.1190$ is far from the threshold. This is consistent with the approach: base 15 was an exploratory observation, not a structural derivation.

9 Limitations

1. QCA is an approach, not a "theory of everything".
2. All successful results reproduce known physics. No new predictions are made; $\Sigma m_\nu = 0.0608$ eV (nb13) is a consistency check with the DESI DR2 upper bound [6], not a prediction.
3. Yukawa couplings are not explained.
4. The interpretation of the "4D operator" is philosophical.
5. **Base 15:** $p_{abs} = 10/84 = 0.1190$**, status not established**; the hypothesis is exploratory, and the framework does not depend on it. The look-elsewhere ratio is 4.76.
6. Neutrinos and constants do not follow the base-15 grid.
7. Three generations: the mechanism is exact in 1D, 2D requires overlap fermions; the number $N = 3$ is input.
8. CP violation: the framework can accommodate it, but the phase δ is input.
9. 2.8% and Koide are not independent (fractional exponents are a consequence of $K = 2/3$).

10. **The look-elsewhere factor 4.76 is a lower bound.** It accounts only for the 39 bases in nb10, not for hypotheses tested in v1–v13 (Koide, SVD, DESI, alternative normalizations). The true multiplicity correction is strictly larger than 4.76, so $p_{abs} = 0.1190$ **is a lower bound, not a full correction**. This makes the base-15 conclusion even less established, not more.

11. Robustness checks (seed, N_{MC}, mean-vs-max aggregation) are not performed in v14.4; the plan is in `nb10_pvalue.md` §6. To be included in v15.

12. Multi-source cosmology is not tested. The approach is neutral to initial conditions.

13. Bounce cosmology and pre-inflationary phases are not tested.

14. Gravity is not tested. The emergence of $1/r^2$ from the QCA lattice is a task for a separate notebook (nb20, planned).

15. SM chiral left-handed doublets are not implemented. Overlap fermions (nb18 Part C) provide the technology, but the specific embedding of $SU(2)_L$ doublets is not done.

10 Conclusion

The "structure and data" approach is confirmed at 8 independent levels of structure (nb01–06, 08, 18), 4 consistency checks with quantum mechanics (nb12, 15–17), 5 negative controls (nb07, 11, 14, 18A, 19 T0–T2), and 4 data classifications (nb09, 10, 13, 19 T4–T5).

Main result of v14.4. The separation of structure and data does not depend on the base-15 hypothesis.

- **8 independent derivations of structure** (Dirac, U(1), SU(2), mass-independence, Higgs, honeycomb lattice, 3D wall, 3 generations) — derived from QCA.
- **4 checks of Class B** (CPT, $c = 1$, entanglement, Pauli) — consistent with quantum mechanics.
- **5 negative controls of Class C** — the classification is non-trivial.
- **4 data classifications** (quarks, base scan, neutrinos, CP phase) — parameters are not derived.
- **Base 15** — an exploratory observation, **not established** ($p_{abs} = 10/84 = 0.1190$, lower bound; look-elsewhere ratio 4.76). The framework does not depend on it.

The thesis "structure is derivable, data is input" is **falsifiable**: the three criteria in Section 8 define the conditions for refutation. None of them has been realized. The approach is neither a "theory of everything" nor the simulation hypothesis; it is a classification of what follows from local unitarity and what must be given from outside.

References

[1] Y. Koide, "New Formula for the Cabibbo Angle and Composite Quarks and Leptons," Phys. Rev. Lett. **47**, 1241 (1981).

[2] P. H. Ginsparg and K. G. Wilson, "A remnant of chiral symmetry on the lattice," Phys. Rev. D **25**, 2649 (1982).

[3] R. Jackiw and P. Rossi, "Zero modes of the vortex-fermion system," Nucl. Phys. B **190**, 681 (1981).

[4] W. P. Su, J. R. Schrieffer, and A. J. Heeger, "Solitons in Polyacetylene," Phys. Rev. Lett. **42**, 1698 (1979).

[5] R. L. Workman *et al.* [Particle Data Group], "Review of Particle Physics," Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

[6] DESI Collaboration, "Cosmology in Extended Parameter Space with DESI Data Release 2 Baryon Acoustic Oscillations," arXiv:2503.14738 [astro-ph.CO] (2025).

[7] G. M. D'Ariano and P. Perinotti, "Quantum cellular automata and free quantum field theory," Frontiers of Physics **12**, 1 (2017).

[8] A. Bisio, G. M. D'Ariano, P. Perinotti, and A. Tosini, "Weyl, Dirac and Maxwell quantum cellular automata," Found. Phys. **45**, 1203 (2015).

[9] S. Lloyd, "Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos," (Alfred A. Knopf, New York, 2006).